

CSCE 640 QUANTUM ALGORITHMS · MODULE 11 · LECTURE NOTES

Near-Term Hardware

What the current generation can and cannot be expected to do.

MODULE OUTCOMES

- Describe the current hardware landscape and its limits.
- Explain variational algorithms and their appeal.
- State the barren plateau problem.
- Explain the input and output problems.
- Assess a near-term application proposal.

1 What exists

CURRENT DEVICES HAVE TENS TO HUNDREDS OF PHYSICAL QUBITS, ERROR RATES NEAR THE THRESHOLD, AND NO USEFUL ERROR CORRECTION

Superconducting circuits, trapped ions, neutral atoms, and photonic systems all exist as working platforms, with very different trade-offs between gate speed, coherence time, and qubit connectivity — which is Module 04 §3's isolation-versus-control tension resolved differently in each case.

Two-qubit gate error rates are around a fraction of a percent on the best systems — which bounds the useful circuit depth to hundreds of gates before the output is noise, **not the millions that Module 08's algorithms require.**

And connectivity is limited: a circuit written assuming all-to-all coupling has to be compiled with swap networks, which multiplies the gate count — run one of your own circuits through a transpiler and watch it happen (Module 03 §4's exercise).

So the honest statement is: these are real quantum devices and they are not fault-tolerant computers — Preskill's 'NISQ' term names exactly this gap, and the gap rather than either side of it is what this module is about. Specific numbers date quickly; the structure of the limitation does not.

2 Variational algorithms

THE IDEA

keep the quantum circuit shallow, and let a classical optimiser do the hard work.

THE LOOP

- 1 prepare a parameterised state $|\psi(\theta)\rangle$ with a short circuit
- 2 measure an expectation value, for example the energy of a Hamiltonian
- 3 a CLASSICAL optimiser proposes new θ
- 4 repeat until converged

INSTANCES

VQE chemistry ground states (Module 09)
 QAOA combinatorial optimisation

WHY IT APPEALS

shallow circuits fit inside the coherence time,
 and some errors are partly absorbed by
 reoptimising the parameters

WHY IT IS DIFFICULT

see section 3: the optimisation landscape

Shallow circuit, classical optimiser — which fits the hardware, and moves the difficulty into the **optimisation**. That relocation is the whole design idea, and it is a reasonable one: the hardware constraint is depth, and a classical optimiser has no depth limit. Whether the relocated problem is easier is §3's subject, and the answer is currently no (CSCE 669's non-convex optimisation material is the right background here).

FOR RANDOM PARAMETERISED CIRCUITS, GRADIENTS VANISH EXPONENTIALLY IN THE NUMBER OF QUBITS

The expected gradient of the cost function is zero and its variance shrinks exponentially with the qubit count — so the landscape is flat almost everywhere, and the optimiser has nothing to follow: this is the barren plateau phenomenon.

Which means the number of measurements needed to resolve a gradient to useful precision grows exponentially — exactly the scaling the whole method was supposed to avoid, and it arrives through the back door of the optimisation rather than through the circuit.

And it is not a hyperparameter tuning problem — it is a property of sufficiently expressive random circuits, provable from the concentration of measure on the unitary group — **and more expressive ansätze make it worse,** which is an uncomfortable trade since expressiveness is what you wanted.

So the mitigations all restrict expressiveness in some way — problem-informed ansätze, shallow and local cost functions, layerwise training, careful initialisation — and restricting expressiveness is in direct tension with capturing the physics you came for, which is the field's central open question rather than a known fix.

3 Input and output

- **The input problem** — loading a classical dataset of size N into quantum amplitudes generally costs $O(N)$ operations, which destroys any polynomial advantage before the algorithm begins (Module 07 §3). Proposed fixes assume a 'QRAM' whose own cost is rarely accounted for.
- **The output problem** — the result of the computation is a quantum state, and extracting more than a few numbers from it costs exponentially many measurements (Module 02 §2's bottleneck) — so an algorithm that 'produces the solution vector' has not produced anything you can read.
- **The classical baseline** — which must be the best known classical method and frequently is not even checked (Module 13 §3), and the dequantisation literature exists precisely because this was skipped.
- **The assumed hardware** — error rate, connectivity, and whether fault tolerance is assumed, all three stated explicitly, because a result assuming error-free qubits is a mathematical result rather than an engineering one.
- **And the scaling claim** — asserted asymptotically, or measured over a range? Usually the former, on a simulated range far too short to fit a trend to. Input and output together eliminate most proposed quantum machine learning applications — which is a strong claim, and is the one the careful papers in the area make rather than an outsider's scepticism.

4 The honest near-term position

AND THE HONEST NEAR-TERM POSITION

No convincing demonstration of a near-term quantum advantage on a useful problem has been made, as of this course's writing — which is a factual statement about the literature rather than a prediction about the future, and should be re-checked rather than assumed.

And the random-circuit sampling demonstrations are real results — a device sampling from a distribution that classical simulation struggles to reproduce — on problems deliberately chosen to be hard to simulate rather than useful, which is the entire point of the experiment and is sometimes lost in the reporting.

Several of those demonstrations have been substantially reduced by improved classical simulation published afterwards (Module 13 §3's pattern) — which is science working rather than scandal, and is the strongest argument for stating baselines carefully in the first place.

So the near-term honest answer to 'what can a hundred noisy qubits do' is: physics experiments, benchmarks, and algorithm development — which is a perfectly reasonable thing for a technology at this stage to be, and is only disappointing against claims that should not have been made.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Preskill — Quantum Computing in the NISQ era and beyond (free)	§1 — the framing, from the person who named it, and it is admirably careful about what is and is not claimed.
Cerezo et al. — Variational quantum algorithms (free)	§2 — the survey, with the obstacles stated alongside the hopes.
McClean et al. — Barren plateaus in quantum neural network training landscapes (free)	§3 in the original — and the result is cleaner and more damaging than its reception suggested.
Aaronson — Read the fine print (free)	§4's checklist — the input and output problems stated once and for all, and the right thing to hand somebody with a proposal.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Find the current best two-qubit error rate** and compute the usable depth.
- 2 **Transpile one circuit** to a limited-connectivity device and compare gate counts.
- 3 **Implement a small VQE** for a two-qubit Hamiltonian.
- 4 **Plot its optimisation landscape** over two parameters.
- 5 **Increase the qubit count** and watch the gradient variance shrink.
- 6 **Try a problem-informed ansatz** and compare.
- 7 **Estimate the cost of loading** a million-element vector into amplitudes.
- 8 **Estimate the measurements needed** to read out a 2^n -element solution vector.
- 9 **Take one published near-term proposal** and run §4's five checks on it.
- 10 **Find a sampling demonstration** and the classical simulation that followed it.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Describe the current hardware landscape in three facts.
- 2 What bounds the usable circuit depth?
- 3 Why does connectivity matter?

- 4 Describe the variational loop and its two instances.
- 5 Why does it appeal, given the hardware?
- 6 State the barren plateau result and its consequence.
- 7 Why is it not a tuning problem?
- 8 What do the mitigations cost?
- 9 State the input and output problems.
- 10 Give the five checks for a near-term proposal.